Philippine Rice Yield Analytics Report

Data Period: 2000 - 2019

Generated: March 10, 2026

Dataset Summary

Provinces Covered: 83

Regions Covered: 16

Total Records: 12,000

National Average Yield: 3.69 MT/ha

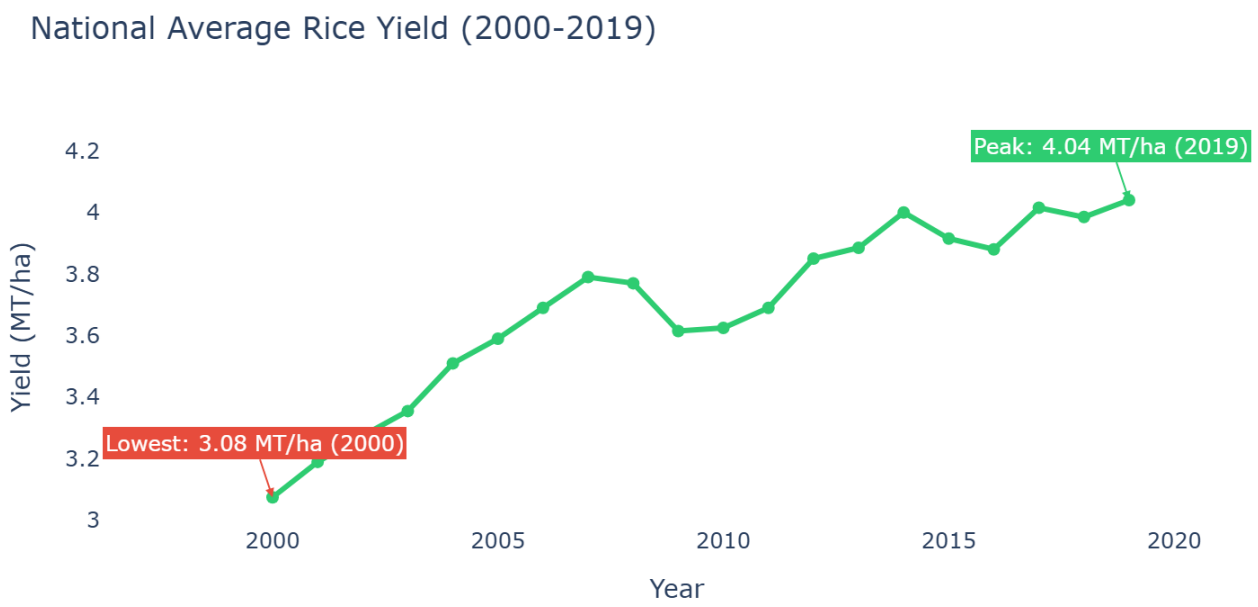
Data Source: Philippine Statistics Authority (PSA)

AI Insights powered by Groq (LLaMA 3.3 70B)

Executive Summary

The Philippine rice production performance from 2000 to 2019 exhibits a notable improvement in yield, increasing from 3.02 to 4.1 MT/ha. The national average yield stands at 3.69 MT/ha, with the best year being 2018 and the worst year being 2000. A significant yield gap persists between irrigated and rainfed areas, with irrigated areas producing an average of 3.59 MT/ha compared to 2.36 MT/ha for rainfed areas. Nueva Ecija emerges as the top-performing province, achieving an impressive average yield of 5.02 MT/ha, highlighting the potential for high-yielding rice production in the country. Overall, the data analysis covering 83 provinces and 16 regions provides valuable insights into the trends and patterns of Philippine rice production, informing strategies for further improvement and development of the sector.

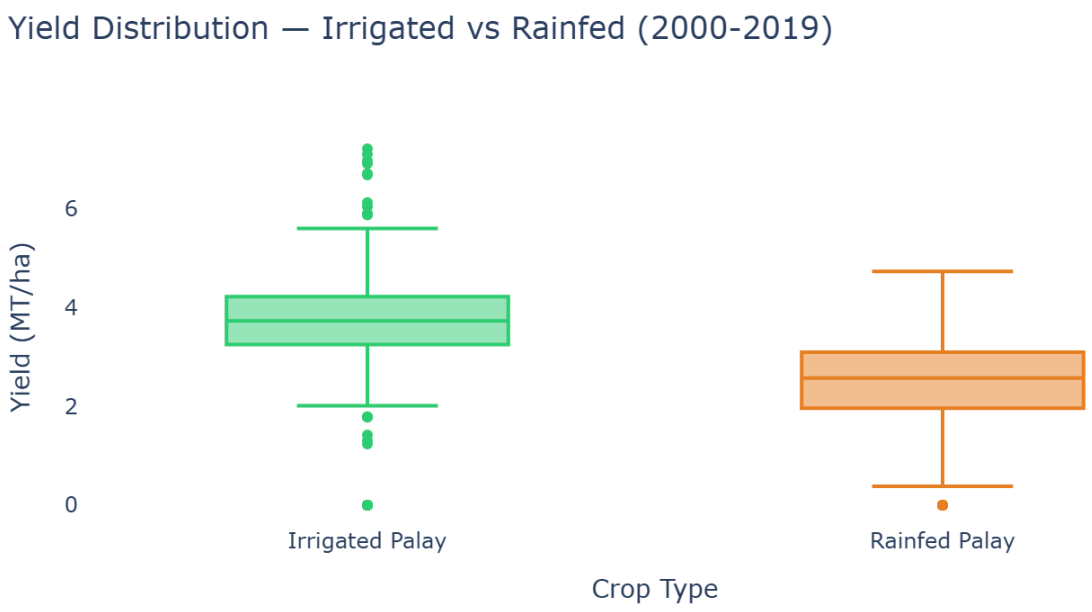
National Yield Trend (2000-2019)



Analysis

The national rice yield trend in the Philippines from 2000 to 2019 exhibits a generally upward trajectory, with an average yield of 3.69 MT/ha over the period. The yield increased significantly from 3.02 MT/ha in 2000 to 4.1 MT/ha in 2018, driven by improvements in farming technologies, irrigation systems, and the adoption of high-yielding rice varieties. However, a notable dip in yield occurred in 2009, which may be attributed to the devastating effects of Typhoon Ondoy and other weather-related disasters that affected rice production during that year. Overall, the trend suggests that the Philippine rice industry has made progress in increasing yield over time, with some fluctuations due to external factors. The highest yield of 4.1 MT/ha in 2018 indicates the potential for further growth and improvement in rice production.

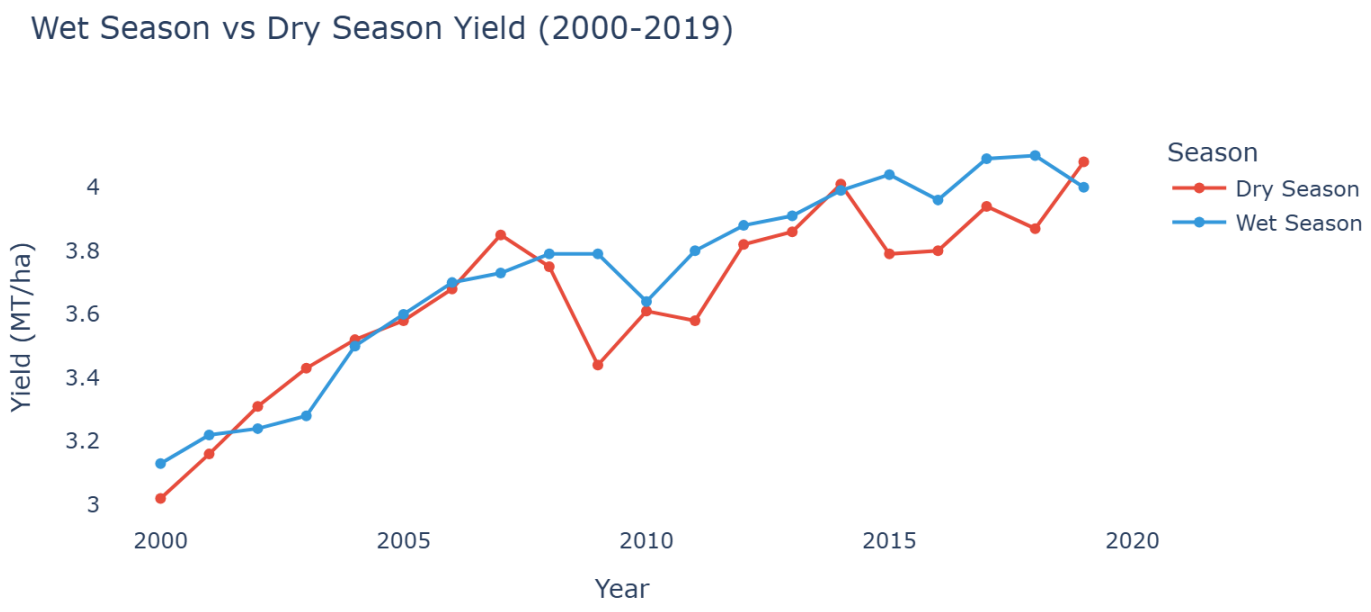
Ecosystem Comparison: Irrigated vs Rainfed



Analysis

The yield difference between irrigated and rainfed rice farming in the Philippines from 2000 to 2019 is substantial, with irrigated farms averaging 3.59 MT/ha and rainfed farms averaging 2.36 MT/ha. This yield gap of 1.23 MT/ha indicates that irrigated farms are significantly more productive, highlighting the importance of reliable water supply in Philippine rice production. The existence of this gap suggests that investments in irrigation infrastructure could greatly boost the country's rice productivity, and policymakers may consider prioritizing initiatives that support the expansion of irrigation systems to rainfed areas. By doing so, the Philippines can increase its overall rice production and reduce its reliance on imports, ultimately enhancing food security. This yield gap also underscores the need for targeted support to rainfed farmers to help them adopt more resilient and productive farming practices.

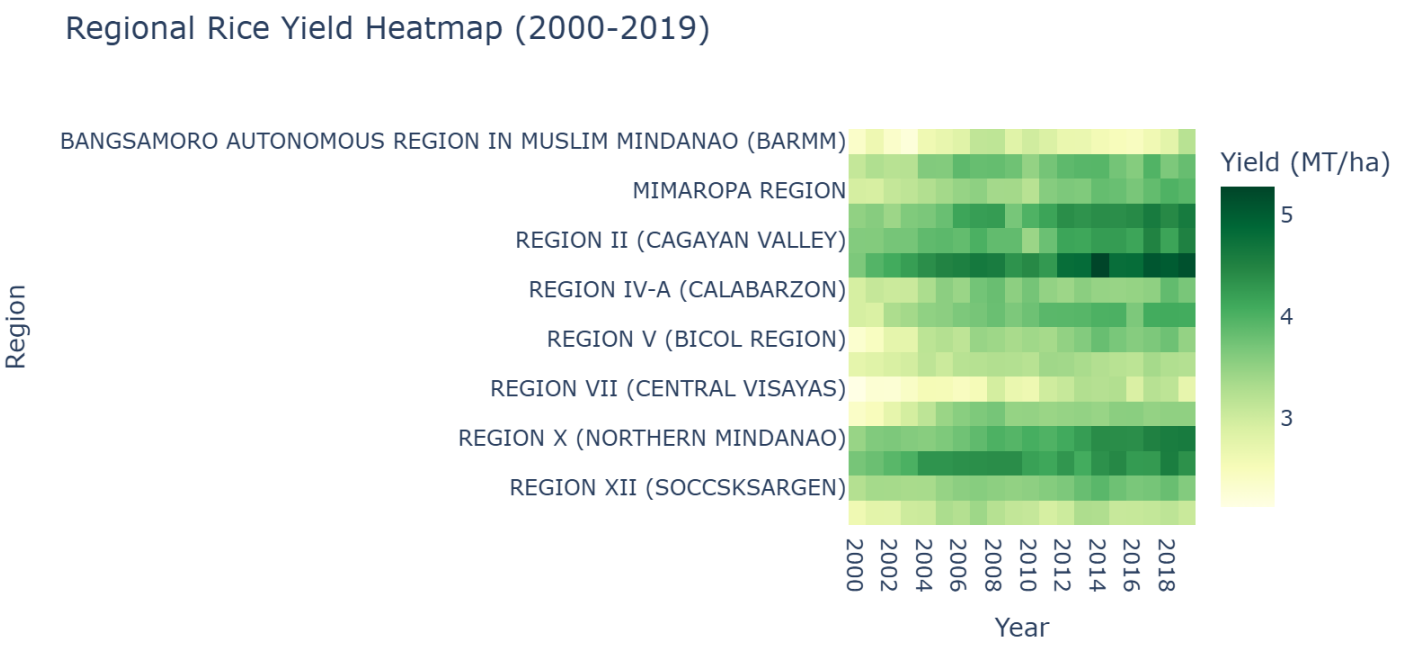
Wet Season vs Dry Season Analysis



Analysis

The Philippines' rice yield patterns from 2000 to 2019 reveal a surprisingly small difference between the wet and dry seasons, with average yields of 3.32 MT/ha and 3.35 MT/ha, respectively. This minimal gap can be attributed to the widespread adoption of irrigation systems, which enable farmers to maintain a consistent water supply during the dry season, thereby mitigating the effects of drought on crop yields. As a result, dry season yields have become more resilient and comparable to those of the wet season, which is typically characterized by abundant rainfall. The role of irrigation in dry season performance is particularly significant, as it allows farmers to optimize crop growth and compensate for the lack of natural precipitation, resulting in yields that are only marginally lower than those achieved during the wet season. Overall, the proximity in yields between the two seasons underscores the importance of irrigation in Philippine rice production.

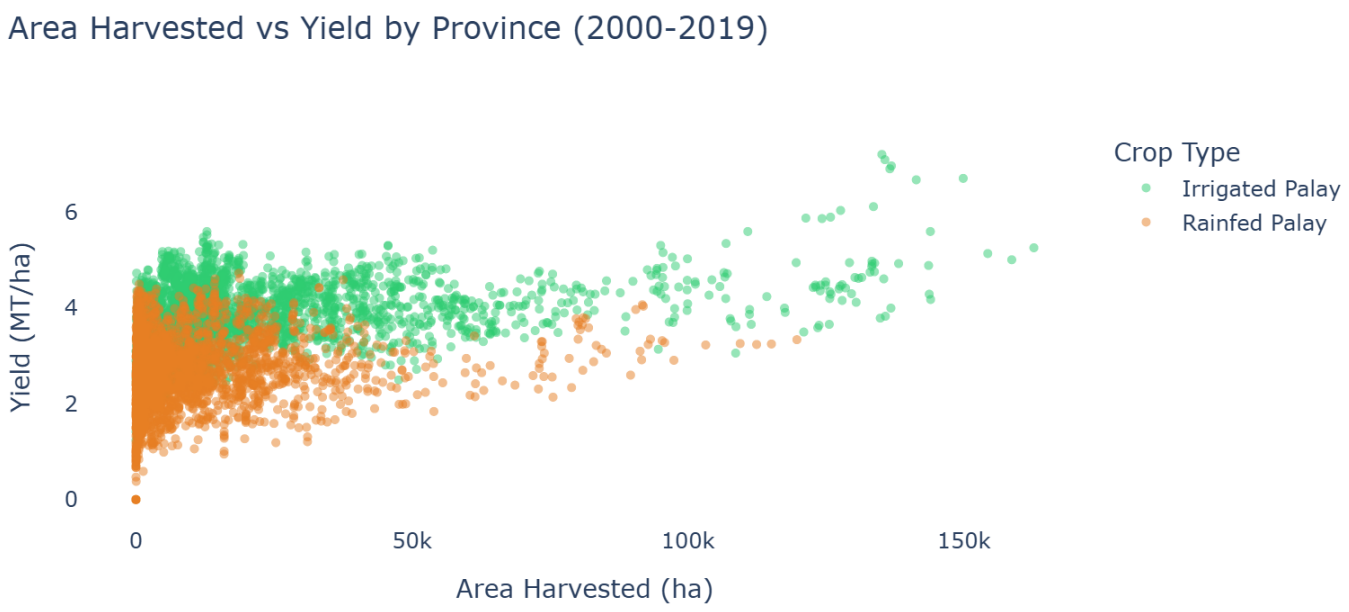
Regional Yield Heatmap



Analysis

The regional rice yield performance across the Philippines from 2000-2019 reveals significant disparities, with top-performing provinces such as Nueva Ecija, Davao del Sur, and Pampanga achieving yields of 5.02, 4.78, and 4.45 MT/ha, respectively. In contrast, provinces like Davao Occidental, Batanes, and Sulu struggled with yields as low as 0.15, 0.43, and 0.73 MT/ha, respectively. Central Luzon's dominance in rice production can be attributed to its fertile soil, favorable climate, and extensive irrigation systems, which enable provinces like Nueva Ecija and Pampanga to consistently achieve high yields. The consistently low-performing regions, on the other hand, are often characterized by limited agricultural infrastructure, rugged terrain, and vulnerability to climate-related stresses, highlighting the need for targeted interventions to improve their rice production capabilities. Overall, these regional disparities underscore the importance of tailored support and investment to enhance the competitiveness of lagging regions.

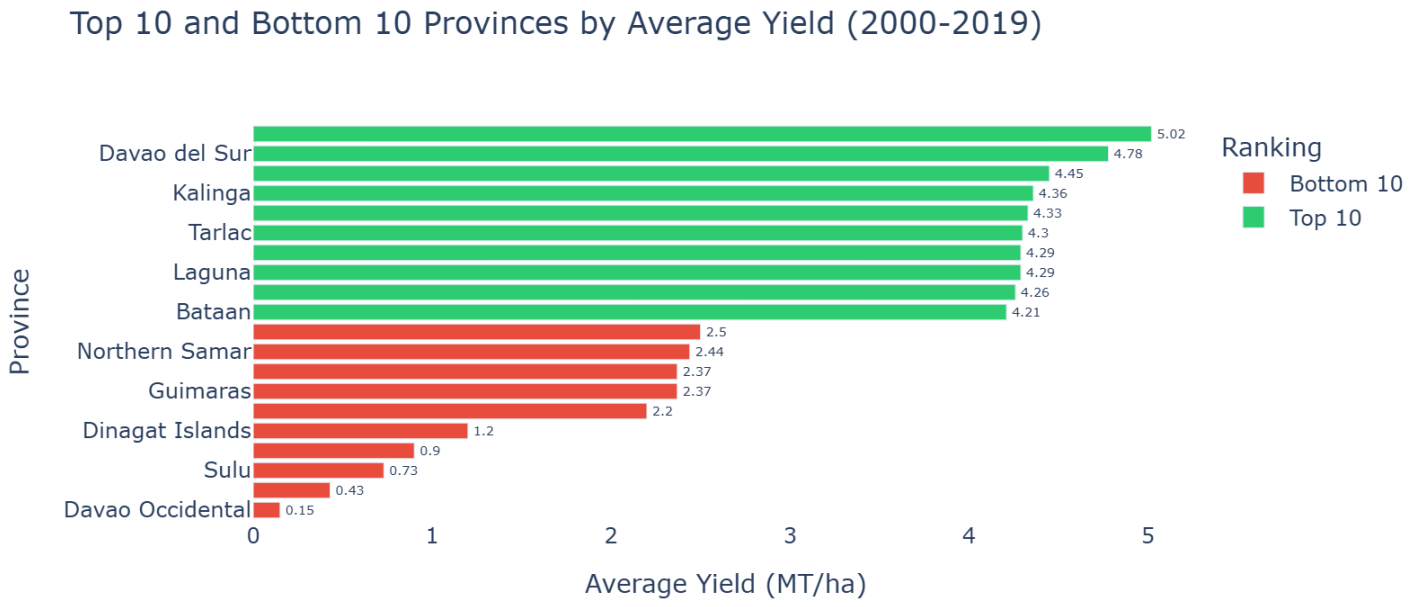
Area Harvested vs Yield



Analysis

The relationship between farm area harvested and rice yield in the Philippines from 2000 to 2019 suggests that farm size does not necessarily drive yield performance. Despite most provinces having a relatively small area harvested, ranging from 0 to 20,000 hectares, the average yield remains modest, with irrigated areas achieving 3.59 MT/ha and rainfed areas at 2.36 MT/ha. Factors such as irrigation, crop management, and soil quality appear to have a more significant impact on yield than farm size, as evidenced by the notable difference in yields between irrigated and rainfed areas. Overall, the data indicates that scale is not the primary determinant of yield performance in Philippine rice farming, and other factors play a more critical role in shaping productivity. As a result, focusing on improving these key factors can lead to more substantial gains in rice yield than simply expanding farm area.

Top & Bottom Provinces by Yield



Analysis

The top-performing provinces for rice yield in the Philippines, led by Nueva Ecija with a yield of 5.02 MT/ha, demonstrate a significant gap from the bottom performers, with Davao Occidental experiencing a substantially lower yield of 0.15 MT/ha. This disparity can be attributed to geographic and infrastructural factors, such as access to irrigation systems and fertile soil, which are more prevalent in the top-performing provinces. The proximity of top-performing provinces to major urban centers and trade routes may also contribute to their success, allowing for more efficient distribution of resources and technology. Overall, the data suggests that a combination of natural resources and infrastructure development plays a crucial role in determining rice yield, with the top performers benefiting from more favorable conditions.